

SAT 8

Math Module 1

Question #

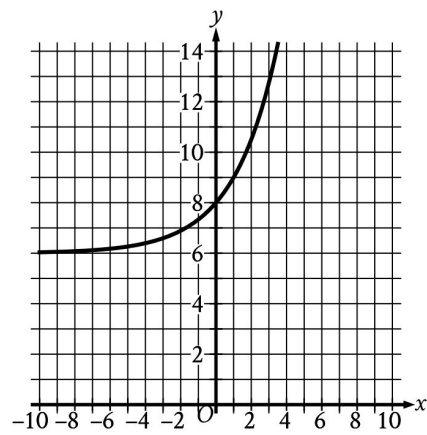
ID

SAT8 M 1.1

f547a8b1

Answer

D



What is the y-intercept of the graph shown?

- A. $(-8, 0)$
- B. $(-6, 0)$
- C. $(0, 6)$
- D. $(0, 8)$

SAT8 M 1.2

a08ma102

The function f is defined by $f(x) = -3x + 60$. What is the value of $f(x)$ when $x = -8$?

D

(A) 49

(B) 52

(C) 57

(D) 84

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Question # ID
SAT8 M 1.3 a08ma103

Answer

D

A producer is creating a video with a length of **70** minutes. The video will consist of segments that are **1** minute long and segments that are **3** minutes long. Which equation represents this situation, where x represents the number of 1-minute segments and y represents the number of 3-minute segments?

☐ (A) $4xy = 70$

☐ (B) $4(x + y) = 70$

☐ (C) $3x + y = 70$

☐ (D) $x + 3y = 70$

SAT8 M 1.4 a08ma104

D

A bowl contains **20** ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by **1** ounce every **4** days. If **9** ounces of water remain in this bowl, for how many days has it been uncovered?

☐ (A) 3

☐ (B) 7

☐ (C) 36

☐ (D) 44

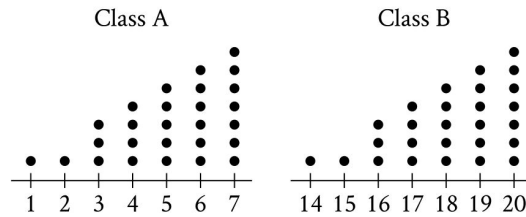
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Question # ID
SAT8 M 1.5 d94018fd

Answer

B



Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

- A. The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.
- B. The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.
- C. The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.
- D. There is not enough information to compare these standard deviations.

SAT8 M 1.6 5805e747

A

Which expression is equivalent to $(7x^3 + 7x) - (6x^3 - 3x)$?

- A. $x^3 + 10x$
- B. $-13x^3 + 10x$
- C. $-13x^3 + 4x$
- D. $x^3 + 4x$

SAT8 M 1.7 a08ma107

C

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

(A) 36

(B) 45

(C) 72

(D) 108

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Question # ID
SAT8 M 1.8 a08ma108

Answer

B

$$3x(x - 4)(x + 5) = 0$$

What is one of the solutions to the given equation?

(A) -4

(B) 0

(C) 3

(D) 5

SAT8 M 1.9 a08ma109

46

A chemist combines water and acetic acid to make a mixture with a volume of **56 milliliters (mL)**. The volume of acetic acid in the mixture is **10 mL**. What is the volume of water, in **mL**, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

SAT8 M 1.10 a08ma110

D

A distance of **354 furlongs** is equivalent to how many feet?
(1 furlong = 220 yards and 1 yard = 3 feet)

(A) 306

(B) 402

(C) 25,960

(D) 233,640

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Question #	ID		Answer
SAT8 M 1.11	6a87902f	$y = 2x + 10$ $y = 2x - 1$ At how many points do the graphs of the given equations intersect in the xy-plane? A. Zero B. Exactly one C. Exactly two D. Infinitely many	A
SAT8 M 1.12	a08ma112	13 is $p\%$ of 25. What is the value of p?	52
SAT8 M 1.13	90bcaa61	The function $f(t) = 60,000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?	410
SAT8 M 1.14	58171b5e	Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time? A. Decreasing exponential B. Decreasing linear C. Increasing exponential D. Increasing linear	C
SAT8 M 1.15	ebbf23ae	A circle in the xy-plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r?	5

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Question # ID
SAT8 M 1.16 33e29881

Answer
B

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$

SAT8 M 1.17 95ed0b69

A

$p = \frac{k}{4j+9}$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

- A. $4j + 9 = \frac{k}{p}$
- B. $4j + 9 = kp$
- C. $4j + 9 = k - p$
- D. $4j + 9 = \frac{p}{k}$

SAT8 M 1.18 48fb34c8

D

$y > 13x - 18$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

- A.

x	y
3	21
5	47
8	86
- B.

x	y
3	26
5	42
8	86
- C.

x	y
3	16
5	42
8	81
- D.

x	y
3	26
5	52
8	91

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Question # ID
SAT8 M 1.19 a08ma119

Answer

0.25

Line ℓ is defined by $3y + 12x = 5$. Line n is perpendicular to line ℓ in the xy -plane. What is the slope of line n ?

SAT8 M 1.20 a08ma120

B

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, d and g are unique positive constants. When the equation is graphed in the xy -plane, how many distinct x -intercepts does the graph have?

(A) 4

(B) 3

(C) 2

(D) 1

SAT8 M 1.21 08d03fe4

C

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A. $f(x) = 50(2)^{x+1}$

B. $f(x) = 80(2)^x$

C. $f(x) = 128(2)^{x-1}$

D. $f(x) = 205(2)^{x-2}$

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Question #	ID
SAT8 M 1.22	841ef26c

Answer

A

$$f(x) = 4x^2 + 64x + 262$$

The function g is defined by $g(x) = f(x + 5)$. For what value of x does $g(x)$ reach its minimum?

- A. -13
- B. -8
- C. -5
- D. -3